Build Fast, Build Simple - SUAS 2025

Schulich Unmanned Aerial Vehicles

I. ABSTRACT

The SUAS 2025 rule set saw a major departure from the heavily rotary-wing favored scoring of previous years. In four major ways, the new rule set is more favorable towards fixed-wing aircraft: lower maximum weight, lower flight floor, no battery swapping, and more forgiving payload drop zones. To take advantage of these changes, Schulich Unmanned Aerial Vehicles (SUAV) chose to pursue a fixed wing design.

Building a fixed wing aircraft under the still strenuous requirements of the current rule set is a daunting challenge. To meet this challenge, SUAV chose to resort to a simple, robust, and effective design, opting out of any choices that would add undue risk to the development cycle. This approach has been successful thus far as it has resulted in SUAV's fastest ever development cycle of a fixed wing aircraft; just over 5 months from concept to maiden flight.

Our competition strategy centers on reliably completing all four payload drops, as they offer the highest scoring potential under the SUAS 2025 rule set. To achieve this, our aircraft was designed to fly slow and stable; ideal for payload delivery and imaging. This design choice aligns with the other competition objectives, including mapping.

Our engineering design prioritized simplicity and systems-level thinking to accelerate integration across all components. This allowed us to quickly repair, manufacture, and iterate on designs.

Mechanically, the design was optimized to carry the full payload over mission distance using the simplest and most maintainable configuration. Electrically, SUAV used a standard setup centered around CubePilot Orange and ArduPlane for flight control. Our Custom Ground Control Station (CGCS) interfaces with ArduPlane to manage autonomous payload drops and imaging, while Mission Planner handles flight navigation.

Testing was structured around isolated development of minimum viable components, gradually integrating them into a complete system. This modular approach enabled validation of systems individually before integration.

II. ACKNOWLEDGEMENTS

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III. COMPETITION STRATEGY

This year, SUAV envisions an aircraft with the speed and endurance to complete all

payload drops and mapping tasks, at the cost of forgoing some of the smaller point categories. Only a fixed-wing design can achieve these performance goals, though it introduces greater challenges for precision payload delivery and imaging tasks. Our experience developing both multirotors and fixed-wings over the past few years has equipped us with the necessary background to confidently pursue a fixed-wing design.

Our aircraft, Orca, is an 11 kg fixed-wing platform designed specifically to carry all required payloads over a 19 km round-trip mission. The aerodynamic design, paired with a slow and stable flight profile, allows for acceptable payload drop accuracy while maintaining the endurance needed to complete the full mission.

Orca has been intentionally designed to qualify for the 100-point category awarded to systems using only sub 100 Wh batteries. This maximizes scoring potential while also simplifying logistics, as these batteries can be transported on commercial airliners without special handling requirements.

SUAV is expected to complete the mission with four people: safety pilot, spotter, ground control station operator, and payload operator. This is not optimal for scoring, as it only scores 100 out of the possible 200 points. However, it has been deemed necessary for ensuring that all aspects of the mission are properly managed. Significant losses, beyond the potential 100 points, may be incurred if the mission is not executed efficiently.

Although up to 50 points are available for small vehicle categories, the size requirements to qualify are extremely restrictive. Given the complexity of the mission, the expected dimensions of a capable vehicle significantly exceed even the largest allowable category. As such, this scoring category was not pursued in the design.

Similarly, points are available for autonomous takeoff and landing; however, these phases are among the highest-risk maneuvers for fixed-wing aircraft and account for the majority of crashes. Given this elevated risk, the team opted for manual landings to ensure greater reliability and mission success.

While the main priority is completing all four payload drops, Orca is also capable of full-area mapping. A Raspberry Pi and global shutter camera are used for image capture, with images transmitted to the ground control station at 4Hz. Flying at higher altitudes increases area coverage, reducing the number of images required to produce a map. After the payload drops, the aircraft follows an efficient route over the area to collect imagery. Testing has shown we can produce high-quality stitched maps quickly and effectively.

IV. DESIGN STRATEGY

SUAV took a systems first approach to the design of Orca. Ease of development, systems integration, manufacturability, and repairability were prioritized above all other factors.

A. Mechanical Design

Orca's design prioritized simplicity to maintain a rapid development pace, an approach most evident in the aircraft's mechanical design.

Features like flaps, dampened landing gear setups, and complex payload deployment mechanisms were intentionally excluded from the design to ensure smooth manufacturing and systems integration.

Aerodynamic Design:

The aerodynamic design aimed to optimize speed for completing the 12 mile mission in under 30 minutes while maintaining high-quality image capture and precise payload delivery. Key constraints were established to guide the design:

- **Payload Mass**
- **Flight Profile**
  - Imaging and Payload Drop Speed
  - Cruise Speed

A linear regression based on previous aircraft component masses estimated the current aircraft's total mass at approximately 11 kg. This linear regression was driven by the expected payload mass.

The flight profile was based on a desired 'low speed' and cruise speed, with the low speed (40 km/h) set comfortably above stall for stable imaging and payload drops, while the cruise speed (60 km/h) was derived from mission time and required distance.

To boost the aerodynamic efficiency of the aircraft, the aspect ratio of the wings was increased to the maximum without hindering logistics.

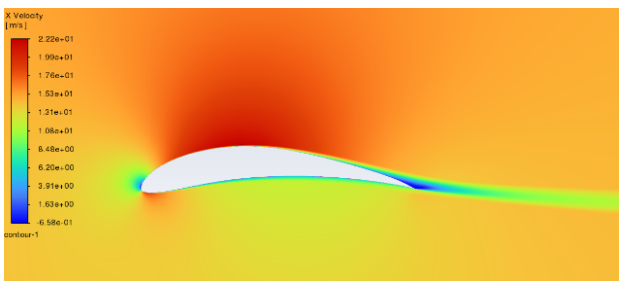

Fig. 1. CFD for the wing airfoil at cruise speed.

Airfoils with a generally high camber were preferred as they would achieve higher lift at slower speeds. A 2D CFD analysis was conducted to assess flow separation risks associated with the high-camber airfoil design. As per Fig. 1, there was no flow separation found. The analysis concluded that the E423 airfoil best suited the needs of the flight profile.

Manufacturing Design:

The team elected to build a carbon composite fuselage using 3D printed molds, and carbon-skinned foam-core wings. These manufacturing methods allow for flexibility in geometry while maintaining the high strength properties of carbon-fiber.

The aim of the custom built fuselage was to ensure that all internal components fit and were readily accessible. To determine the approximate dimensions and skin shape, 3D models of electronics and payload were arranged based on wiring connections and center of gravity placement. Based on feedback from previous years, it was decided that the aircraft must have a removable top to ensure access to avionics. Models of both the skin shape and molds were created in Onshape and Solidworks, as seen in Fig. 2 and Fig. 3.

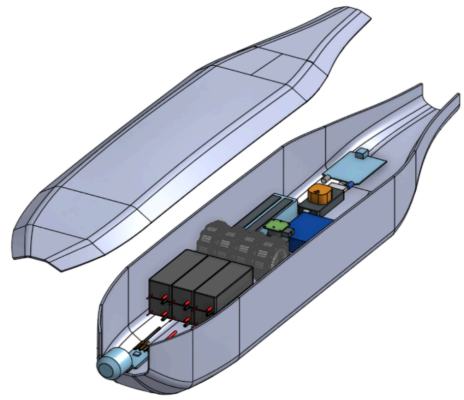

Fig. 2. Fuselage skin shape with electronics (roof detached) in Onshape.

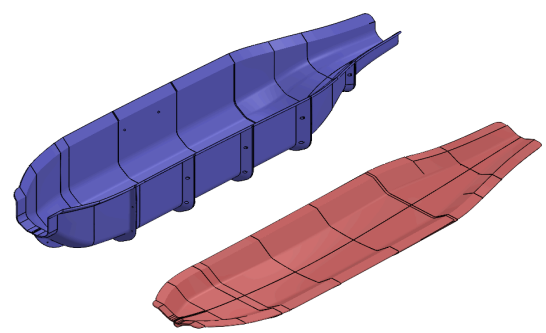

Fig. 3. Roof mold (red) and bottom mold (purple), in Solidworks.

For the aircraft's wings and aerodynamic surfaces, SUAV approached the manufacturing with a focus on efficiency and repeatability, using lightly modified processes from previous aircraft. Custom foam cores for carbon fiber layups were cut out using a hotwire CNC foam cutting machine to achieve the desired airfoil, then laid up in a single layer of 0/90 TeXtreme carbon-fiber. The manufacturing processes for the aerodynamic surfaces were easily replicable to produce backup parts. Excluding painting, it is possible to produce an aircraft wing within a single 8 hour work day.

Payload Design:

A number of options regarding payload deployment were considered, these include but are not limited to: standard parachute, Delayed Parachute Release (DPS), and non parachute (i.e. gliders). Ultimately, a standard parachute design was chosen for its reliability, and streamlined integration.

On Orca, there are four parachute bays. Each bay contains one beacon, its parachute, and a servo to actuate a trap door. See Fig. 4.

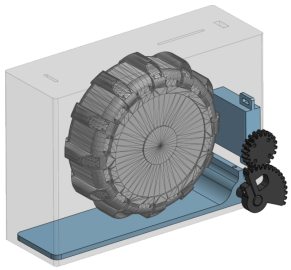

Fig. 4. Single payload bay.

B. Electrical Design

Orca's electrical system is driven by the Orange Cube Pixhawk flight controller, running ArduPlane. The Orange Cube was selected for its modular design and redundant IMUs. It is paired with dual GPS modules operating in blend mode to enhance positional accuracy, along with a pitot tube for airspeed sensing. Two independent Battery Eliminator Circuits (BECs) supply regulated power to the Orange Cube and its servo rail.

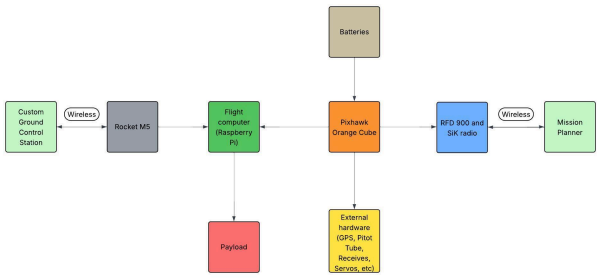

Fig. 5. High level electrical block diagram.

A Raspberry Pi 5 with 8 GB of RAM serves as the mission computer. It directly interfaces with a Raspberry Pi Global Shutter Camera Module, chosen for its ability to capture clear images during high-speed flight. Using a native Pi camera streamlines hardware integration and software control. The camera is equipped with a 2.8-12 mm industrial telephoto varifocal lens, offering an effective balance between zoom and focus.

Power is provided by commercially available, off-the-shelf batteries. Each battery is rated below 100 Wh to comply with transport safety regulations. A 96 Wh battery powers the mission computer for over four hours of operation. Propulsion and servo actuation are handled by a 2S3P Li-Po battery configuration.

To ensure robust and redundant telemetry and communication, a multi-link system redundancy is used. A RocketM5 (5 GHz) provides high-throughput data transfer with our Custom Ground Control Station (CGCS). Images are transmitted to the CGCS at 4 Hz over this link. A SiK radio paired with an RFD900 module (900 MHz) supports communication with Mission Planner, while an FrSky X9D L9R system (2.4 GHz) offers manual RC control. This enables fully offline operation, independent of internet connectivity. This setup has been used successfully in past SUAS competitions without telemetry issues.

C. Software Design

Orca's software system was built around three core principles: simplicity, reliability, and efficiency. There are three main software components:

1. Mission Planner (an open-source GCS)
2. Web-based Custom Ground Control Station (CGCS)
3. Onboard mission computer

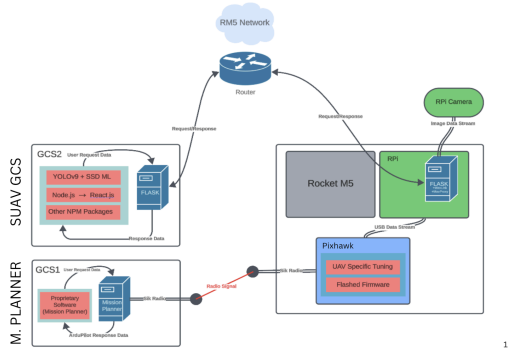

Fig. 6. Software architecture and communications.

Mission Planner is used by the GCS operator to manage flight systems and mission waypoint navigation. The CGCS interfaces with the onboard mission computer to handle advanced capabilities like imaging, mapping, and payload delivery. Through this interface, operators can dynamically plan missions, trigger payload drops, view and command flight data, initiate map processing, and control the imaging system. All target detections are displayed on the map in the CGCS, with color-coding used to identify outliers and eliminate false positives.

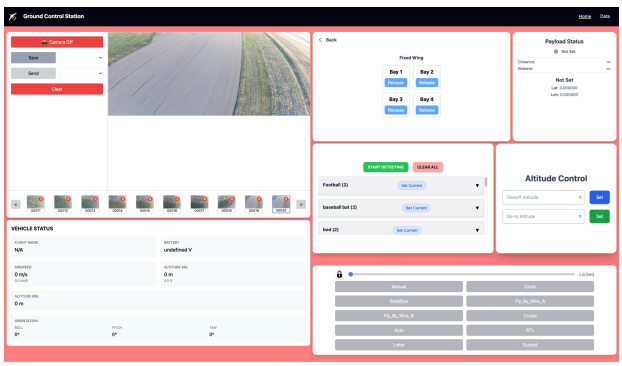

Fig. 7. Custom Ground Control Station.

For robust communication, the CGCS and mission computer communicate via RESTful APIs over TCP, while the flight controller uses the MAVLink protocol to interface with the mission computer and Mission Planner.

SUAV's design prioritizes onboard computation to reduce telemetry reliance and improve control loop performance. The system uses an event-driven architecture, listening for commands from the CGCS to interface with the imaging system, payload servos, and flight controller. With ArduPlane, MAVLink, and MAVProxy, the system is capable of autonomous decision-making based on live data and mission status.

Images captured during flight are GPS-tagged and sent to the CGCS, where a YOLOv11-based AI model (running in a Docker GPU container) detects objects within the drop zone. A custom refinement model adjusts for lens distortion and vehicle attitude at the time of capture to output precise GPS coordinates for each target.

Once a target is confirmed, the system automatically uploads a waypoint mission consisting of:

- ➔ **Entry waypoint:** Aligns the fixed-wing UAV with zero pitch and roll for stable approach.
- ➔ **Target waypoint:** Positioned at the detected target; the payload is released when the UAV is within 20 m at low speed and altitude.

For mapping, the CGCS integrates open source OpenDroneMap (ODM) software to generate high-resolution ortho maps from aerial imagery. Optimized for this system, ODM can process over 400 images in under 2 minutes, producing outputs comparable to Google Maps.

V. TESTING STRATEGY

As with any complex systems engineering challenge, it is best to begin with a minimum viable product, then iteratively test and add more functionality until all components are integrated. Isolated systems testing was completed by the end of February to leave sufficient time for integration and full system testing ahead of the flight readiness deadline in late April.

Upon completion of the manufacturing of the aircraft, the following tests were performed:

Wing Loading Test:

The aircraft was loaded with its maximum weight, then picked up from the wing tips. This loaded the wings in a manner similar to what is experienced during 3-4 G load, beyond the maximum that this aircraft would see in flight.

Landing Gear Drop Test:

To simulate hard landings, the aircraft was again loaded to its maximum weight, then dropped. Using data from previous aircraft, it is known that a 5.5 m/s (vertical speed)

touchdown is a rough landing. This was calculated to correlate to a 1.5 m drop.

Propulsion Systems Testing:

The aircraft's batteries, electronic speed controller (ESC), motor, and propeller were mounted on a thrust testing stand and subjected to rigorous load testing to test reliability. The thrust stand was run at 100% throttle for four minutes. This is much higher than our average flight throttle of 60%. Then the test was repeated at a realistic throttle level to confirm thrust is sufficient for Orca and to assess long-term battery performance under load.

Fig. 8. Thrust stand testing.

Payload Testing:

Payload failure—specifically, a partial release where the parachute remains attached to the aircraft—was identified as a significant risk, as it would almost certainly result in a crash. To mitigate this, the payload was assembled and tested independently on the ground before integration and aerial testing.

Telemetry Testing:

All telemetry tests were conducted on the ground, starting with students side by side, then walking apart. The RocketM5 achieved 40.5 Mbps with only 1.8% packet loss at 1.1 km. Both the SiK Radio and RC links remained stable beyond 500 m and visual line of sight.

Software Testing:

The software testing framework includes unit tests and feature-level testing. Given the mission-critical nature of the system, all functionality not covered by automated testing underwent verification through either manual smoke testing or Formal Qualification Testing (FQT). Various tools were used for testing the software including, Selenium for the GCS UI, PyTest for the backend logic, and Software-in-the-Loop (SIL) for simulating full mission workflows. Hardware-in-the-Loop (HIL) testing was used to validate real-time behaviour between the flight controller and mission computer.

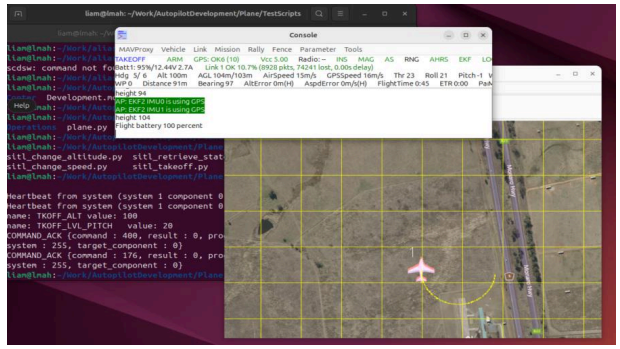

Fig. 9. Software-in-the-Loop (SIL) testing.

Full System Testing:

The full system testing protocol prioritized safety through a phased approach from manual control to full autonomy. Initial manual flights evaluated aircraft stability, vibration characteristics, and maneuverability. Following successful validation, testing progressed to assisted flight modes including FBWA and AutoTune, culminating in autonomous mission execution.

After verifying assisted flight mode safety, imaging tests commenced to validate object detection and mapping capabilities. Autonomous flight reliability via Mission Planner enabled testing of custom mission uploads based on detection outputs from the Custom Ground Control Station (CGCS).

Payload deployment testing began with manual triggering during initial flight phases. Upon confirming payload system reliability, testing advanced to complete autonomous end-to-end drop missions using custom-generated waypoints.

VI. REFERENCES

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VII. APPENDIX A: TEST PLAN AND RESULTS

Each system followed an isolated test plan to verify and validate the functionality provided by the end system.

The propulsion test was classified as high-risk and safety-critical, and underwent extensive testing to ensure flight readiness. The following test was conducted to validate system reliability:

- ➔ **Max Throttle Test:** The motor was operated at **100% throttle for four minutes**. System behaviour was closely monitored to ensure smooth operation, with no unusual sounds or signs of overheating. Current draw was cross-checked against ESC specifications to confirm it remained within safe limits. All tests passed successfully. The maximum current spike recorded was 120.2 A, providing a **29.8 A safety margin** within the 150 A ESC current rating.
- ➔ **Thrust Stand Endurance Test:** Endurance testing was performed at 60% throttle to replicate typical cruise conditions. Starting from a full charge (25.2 V), the battery was discharged to 10% capacity (22.12 V) resulting in a continuous runtime of 51.3 minutes. At a cruise speed of 17 m/s, this translates to an **estimated range of 58.15 km.**
- ➔ **Aerial Endurance Test:** Data was cross-validated with data from a flight covering 5.24 km, during which 8% of battery was consumed (including takeoff and landing). This suggests an **estimated range of 65.5 km**, which exceeds the theoretical value due to a higher observed cruise speed of approximately 19.5 m/s.

Software requirements for the Custom Ground Control Station (CGCS) and mission computer are verified through a combination of feature testing and manual smoke testing. Given the mission-critical nature of the system, all tests must pass before a software version is approved for flight testing. These tests trace requirements and also serve as regression tests to prevent unintended behaviour.

Wherever possible, automated software tests were used to verify functionality. When automation wasn't feasible, manual smoke tests were performed, often using Software-in-the-Loop (SITL/SIL) simulations. For features requiring physical components, Hardware-in-the-Loop (HIL) testing was conducted using legacy hardware from previous years.

CGCS SOFTWARE FEATURE TESTS		
Requirement	**Testing Platform**	**Expected Result**
Image Panel - Camera Toggle	HIL	When toggled on, the camera button turns green and captures images with geodata at 4 Hz. When toggled off, the button turns red and no images are taken.
Image Panel - Viewing	Automated Software Test	Selecting an image from the photo panel displays it in the viewer.
Image Panel - Picture Scrolling	Automated Software Test	Clicking the side arrows scrolls through captured images.
Image Panel - Clear Button	Automated Software Test	Pressing the clear button deletes all locally stored images.
Image Panel - Manual Coordinate Selection and Save	Automated Software Testing	Clicking a point in an image and saving it stores the coordinates in a JSON file.
Image Panel - Manual Coordinate Mission Upload	Manual Smoke Test, SIL	Clicking the send button uploads a mission to the selected drop point.
Payload Status Panel	Manual Smoke Test, SIL	When a payload drop mission is uploaded and the vehicle is in AUTO mode, the panel updates at 1 Hz.
Vehicle Status Panel	Manual Smoke Test, SIL	The panel updates at 1 Hz with live vehicle telemetry.
Altitude Control Panel	Manual Smoke Test, SIL	Pressing the “SET” button in GUIDED mode commands the vehicle to fly at the specified altitude.
Flight Mode Panel - Lock Function	Automated Software Testing	The flight mode cannot be changed until the lock slider is dragged.
Flight Mode Panel - Mode Change	Manual Smoke Test, SIL	Clicking a flight mode button changes the vehicle mode accordingly.
Payload Panel	HIL	Pressing the “Release” button opens the payload bay for 3 seconds, then closes it.
AI Panel - Start Detecting Button	Manual Smoke Test	Pressing “Start Detecting” processes all captured images for object classification.
AI Panel - Clear All Button	Automated Software Testing	Pressing “Clear All” removes all cached object detections.
AI Panel - Set Current	Manual Smoke Test, SIL	Pressing “Set Current” uploads a payload drop mission to the flight controller.
AI Panel - Dropdown	Manual Smoke Test	Clicking the dropdown for a detected object displays all associated coordinate estimates. Outliers are highlighted with a red border.

Autonomous Payload Deployment	Manual Smoke Test, SIL	A mission upload creates an entry and target waypoint. The aircraft maintains < 5°, roll/pitch and 18 m altitude. The payload release is triggered ~20 m from the target waypoint.
Data Panel - UI	Automated Software Testing	Target pixel coordinates are shown in the dropdown menu.
Data Panel - Clear	Automated Software Testing	Pressing “Clear” removes the selected target’s data.
Data Panel - Delete Coordinate	Automated Software Testing	Pressing the “X” next to a coordinate deletes the corresponding saved data.
Data Panel - Map Tab	Manual Software Testing	Detected targets are drawn on the map, with outliers highlighted in yellow, orange, or red based on severity.